Concurrent Kleene Algebra: Free Model and Completeness

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Introduction

Kleene Algebra models *program flow*.

- abort (0) and skip (1)
- atomic actions (a, b, ...)
- non-deterministic choice ($+$)
- sequential composition ($\cdot$)
- indefinite repetition (*)

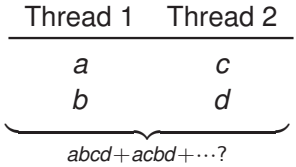
$$(e + f)^* \equiv_{\mathsf{KA}} e^* \cdot (f \cdot e^*)^*$$

Introduction

Thread 1	Thread 2
a	c
b	d

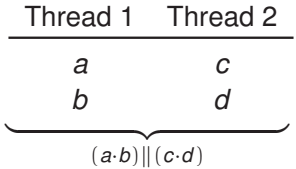
How do we model concurrent composition?

Introduction



Interleaving is a stop-gap: concurrency information lacking from traces.

Introduction



Concurrent KA[1] adds *parallel composition* (||)

[1] Hoare, Möller, Struth, and Wehrman 2009.

Introduction

KA is well-studied:

- Decision procedures [Hopcroft and Karp 1971; Bonchi and Pous 2013]
- Automata, coalgebra [Kleene 1956; Brzozowski 1964; Silva 2010]
- Free model, completeness [Salomaa 1966; Conway 1971; Kozen 1994]

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CKA is a work in progress:

- Decision procedures [Brunet, Pous, and Struth 2017]
- Automata [Lodaya and Weil 2000; Jipsen and Moshier 2016]
- Free model, completeness [Gischer 1988; Laurence and Struth 2014]

See also [K., Brunet, Luttik, Silva, and Zanasi 2017].

Introduction

Lemma (Kozen 1994)

The axioms for KA *are complete for equivalence:*

$$e \equiv_{\mathsf{KA}} f \iff [\![e]\!]_{\mathsf{KA}} = [\![f]\!]_{\mathsf{KA}}$$

$[\![-]\!]_{\mathsf{KA}}$ *is the regular language interpretation of* e.

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Question

Can we find axioms for CKA *that are complete for equivalence? That is,*

$$e \equiv_{\mathsf{CKA}} f \overset{?}{\iff} [\![e]\!]_{\mathsf{CKA}} = [\![f]\!]_{\mathsf{CKA}}$$

$[\![-]\!]_{\mathsf{CKA}}$ *is a generalized regular language interpretation of* e.

Caveat auditor

Completeness for CKA is also shown in [Laurence and Struth 2017]; c.f.

`https://arxiv.org/abs/1705.05896`

Our method differs, because it...

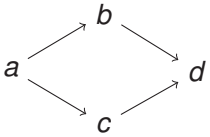
- ...is fully syntactic
- ...uses fixpoints instead of congruences
- ...is explicitly constructive

We do owe part of our method to op. cit.

Preliminaries

- Pomset: “word with parallelism”

$$a \cdot (b \parallel c) \cdot d =$$



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$$a \cdot (b \parallel c) \cdot d = \quad a \;\begin{matrix} \nearrow b \searrow \\ \searrow c \nearrow \end{matrix}\; d$$

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  - $\mathcal{U} \cdot \mathcal{V} = \{U \cdot V : U \in \mathcal{U}, V \in \mathcal{V}\}$
  - $\mathcal{U} \parallel \mathcal{V} = \{U \parallel V : U \in \mathcal{U}, V \in \mathcal{V}\}$

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  - $\mathcal{U} \parallel \mathcal{V} = \{U \parallel V : U \in \mathcal{U}, V \in \mathcal{V}\}$
- Kleene star: $\mathcal{U}^* = \bigcup_{n<\omega} \mathcal{U}^n$

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BKA *semantics* is given by $[\![-]\!]_{\text{BKA}} : \mathcal{T} \to 2^{\text{Pom}_\Sigma}$.

$$[\![0]\!]_{\text{BKA}} = \emptyset \qquad [\![e + f]\!]_{\text{BKA}} = [\![e]\!]_{\text{BKA}} \cup [\![f]\!]_{\text{BKA}} \qquad [\![e^*]\!]_{\text{BKA}} = [\![e]\!]^*_{\text{BKA}}$$
$$[\![1]\!]_{\text{BKA}} = \{1\} \qquad [\![e \cdot f]\!]_{\text{BKA}} = [\![e]\!]_{\text{BKA}} \cdot [\![f]\!]_{\text{BKA}}$$
$$[\![a]\!]_{\text{BKA}} = \{a\} \qquad [\![e \parallel f]\!]_{\text{BKA}} = [\![e]\!]_{\text{BKA}} \parallel [\![f]\!]_{\text{BKA}}$$

Preliminaries

Axioms for BKA :

$$e + 0 \equiv_{\mathsf{BKA}} e \qquad e \cdot 1 \equiv_{\mathsf{BKA}} e \equiv_{\mathsf{BKA}} 1 \cdot e \qquad e \cdot 0 \equiv_{\mathsf{BKA}} 0 \equiv_{\mathsf{BKA}} 0 \cdot e$$

$$e + e \equiv_{\mathsf{BKA}} e \qquad e + f \equiv_{\mathsf{BKA}} f + e \qquad e + (f + g) \equiv_{\mathsf{BKA}} (f + g) + h$$

$$e \cdot (f \cdot g) \equiv_{\mathsf{BKA}} (e \cdot f) \cdot g \qquad e \cdot (f + g) \equiv_{\mathsf{BKA}} e \cdot f + e \cdot h \qquad (e + f) \cdot g \equiv_{\mathsf{BKA}} e \cdot g + f \cdot g$$

$$1 + e \cdot e^* \equiv_{\mathsf{BKA}} e^* \qquad e \cdot f + g \leq_{\mathsf{BKA}} f \implies e^* \cdot g \leq_{\mathsf{BKA}} f$$

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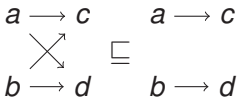
Theorem (Laurence and Struth 2014)

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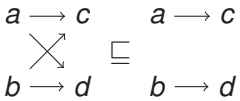
Preliminaries

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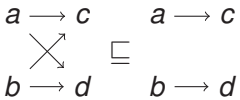
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- *Closure* under pomset subsumption: $\mathcal{U}{\downarrow} = \{U' \sqsubseteq U : U \in \mathcal{U}\}$

  $\mathcal{U}{\downarrow}$: all "sequentialisations" of pomsets in $\mathcal{U}$.

Preliminaries

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- For instance

$$[\![a \parallel b]\!]_{\mathrm{BKA}} = \{a \parallel b\}$$
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- Axioms to build $\equiv_{\mathsf{CKA}}$: all axioms for $\equiv_{\mathsf{BKA}}$, as well as the *exchange law*:

$$(e \parallel f) \cdot (g \parallel h) \leq_{\mathsf{CKA}} (e \cdot g) \parallel (f \cdot h)$$

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Lemma (Hoare, Möller, Struth, and Wehrman 2009)

The axioms of CKA are sound for equivalence, i.e.,

$$e \equiv_{\mathsf{CKA}} f \implies [\![e]\!]_{\mathsf{CKA}} = [\![f]\!]_{\mathsf{CKA}}$$

Preliminaries

Lemma (Kozen 1994)

Let M be an n-by-n matrix over $\mathcal{T}$, and $\vec{b}$ an n-dimensional vector over $\mathcal{T}$.

The inequation $M \cdot \vec{x} + \vec{b} \leq_{\mathsf{KA}} \vec{x}$ admits a unique least solution (with respect to $\leq_{\mathsf{KA}}$).

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- In fact, the solution is the same in both systems!
- We use this as a device to find specific terms later on.

Closure

Definition

Let $e \in \mathcal{T}$; a *closure* of e is a term $e{\downarrow}$ such that

1. $e{\downarrow} \equiv_{\mathrm{CKA}} e$
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If every term e has a closure $e{\downarrow}$, then $[\![e]\!]_{\mathsf{CKA}} = [\![f]\!]_{\mathsf{CKA}}$ implies $e \equiv_{\mathsf{CKA}} f$.

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Proof.

Observe that $[\![e{\downarrow}]\!]_{\mathsf{BKA}} = [\![f{\downarrow}]\!]_{\mathsf{BKA}}$, and therefore $e \equiv_{\mathsf{CKA}} e{\downarrow} \equiv_{\mathsf{BKA}} f{\downarrow} \equiv_{\mathsf{CKA}} f$. □

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Lemma

If e, f have closures $e{\downarrow}$ and $f{\downarrow}$ respectively, then

1. $e{\downarrow} + f{\downarrow}$ *is a closure of* $e + f$
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One case remains: parallel composition.

Induction hypothesis: for $e \in \mathcal{T}$, we assume that:

- If f is a strict subterm of e, we can construct $f\downarrow$.
- If $|f| < |e|$ we can construct $f\downarrow$.[2]

[2] $|e|$ is the nesting level of e w.r.t. $\|$

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Sketch: given $e \parallel f$, apply exchange law syntactically, “in the limit”.

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For instance: if $e = a \cdot b$ and $f = c \cdot d$:

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- $(1 \parallel c) \cdot ((a \cdot b) \parallel d) \leqq_{\mathsf{CKA}} e \parallel f$ $\qquad (e = 1 \bullet a \cdot b,\ f = c \bullet d)$
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- $\ldots$

Goal: find enough of these terms to cover all pomsets in $[\![e \parallel f]\!]_{\mathsf{CKA}}$.

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- How to split terms e and f into heads and tails? ☞ splicing relations
- What to do about recursion? For instance, ☞ fixpoints of inequations

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Closure

Definition

Let $e \in \mathcal{T}$. We define $\nabla_e \subseteq \mathcal{T} \times \mathcal{T}$ as the smallest relation such that

$$\frac{}{1 \ \nabla_1 \ 1} \qquad \frac{}{a \ \nabla_a \ 1} \qquad \frac{}{1 \ \nabla_a \ a} \qquad \frac{}{1 \ \nabla_{e^*} \ 1} \qquad \frac{\ell \ \nabla_e \ r}{\ell \ \nabla_{e+f} \ r} \qquad \frac{\ell \ \nabla_f \ r}{\ell \ \nabla_{e+f} \ r}$$

$$\frac{\ell \ \nabla_e \ r}{\ell \ \nabla_{e \cdot f} \ r \cdot f} \qquad \frac{\ell \ \nabla_f \ r}{e \cdot \ell \ \nabla_{e \cdot f} \ r} \qquad \frac{\ell_0 \ \nabla_e \ r_0 \qquad \ell_1 \ \nabla_f \ r_1}{\ell_0 \parallel \ell_1 \ \nabla_{e \parallel f} \ r_0 \parallel r_1} \qquad \frac{\ell \ \nabla_e \ r}{e^* \cdot \ell \ \nabla_{e^*} \ r \cdot e^*}$$

Lemma

Let $e \in \mathcal{T}$ and $U \cdot V \in [\![e]\!]_{\mathrm{WCKA}}$; there exist $\ell \ \nabla_e \ r$ such that $U \in [\![\ell]\!]_{\mathrm{CKA}}$ and $V \in [\![r]\!]_{\mathrm{CKA}}$.

Closure

Suppose that for all $g, h \in \mathcal{T}$, we have that $X_{g \parallel h}$ is a closure of $g \parallel h$.

Then we find

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Lemma

Continuing this, we get a finite system of inequations $\langle M, \vec{b} \rangle_{e \parallel f}$.

Closure

Theorem

Let $e \otimes f$ be the least solution to $X_{e \| f}$ in $\langle M, \vec{b} \rangle_{e \| f}$. Then the following hold:

1. $e \otimes f \equiv_{\mathrm{CKA}} e \parallel f$
2. $[\![e \otimes f]\!]_{\mathrm{BKA}} = [\![e \parallel f]\!]_{\mathrm{CKA}}$

In other words, $e \otimes f$ is a closure of $e \parallel f$.

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If $e \in \mathcal{T}$, then we can compute a term $e{\downarrow}$ that is a closure of e.

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Theorem

If $e \in \mathcal{T}$, then we can compute a term $e{\downarrow}$ that is a closure of e.

Corollary

Let $e, f \in \mathcal{T}$ be such that $[\![e]\!]_{\mathrm{CKA}} = [\![f]\!]_{\mathrm{CKA}}$; then $e \equiv_{\mathrm{CKA}} f$.

Conclusion

- Axiomatised equality of *closed, series-rational pomset languages*.
- Results establishes these as the carrier of the free CKA.
- Extends half of earlier Kleene theorem: terms to pomset automata.
- We also obtain a novel (but inefficient) decision procedure.

Further work

- Explore coalgebraic perspective:
  - Efficient equivalence checking through bisimulation?
  - Can completeness be shown coalgebraically?
- Add “parallel star” operator — closure method does not apply.
- Endgame: lift results to KAT, then NetKAT.

Thank you for your attention



Implementation: `https://doi.org/10.5281/zenodo.926651`.

Draft paper: `https://arxiv.org/abs/1710.02787`.